Rural Sociology 88(1), 2023, pp. 71–107
DOI: 10.1111/ruso.12469

The Farming Question: Intergenerational Linkages, Gender and Youth Aspirations in Rural Zambia☆

Oluwafemi Ogunjimi
Division of Social and Institutional Change in Agricultural Development, Hans-Ruthenberg-Institute of Agricultural Science in the Tropics
University of Hohenheim

Thomas Daum
Division of Social and Institutional Change in Agricultural Development, Hans-Ruthenberg-Institute of Agricultural Science in the Tropics
University of Hohenheim

Juliet Kariuki
Division of Social and Institutional Change in Agricultural Development, Hans-Ruthenberg-Institute of Agricultural Science in the Tropics
University of Hohenheim

ABSTRACT With agriculture considered key to generating jobs for Africa's growing population, several studies have explored youth aspirations toward farming. While many factors explaining aspirations have been well studied, little is known about the actors' shaping aspirations. We developed a novel framework that focuses on the factors and actors shaping the formation and actual aspirations of rural youth and applied a unique "whole-family" approach based on mixed-methods data collection from adolescents (boys and girls) and corresponding adults. We applied this approach in rural Zambia, collecting data from 348 adolescents and adults in 87 households. The study finds that parents strongly shape youth aspirations—they are much more influential than siblings, peers, church, and media. Male youth are more likely to envision farming (full or part-time) than female youth. The male preference for farming reflects their parent's aspirations and is reinforced by the patriarchal system of land inheritance. Parents' farm characteristics, such as degree of mechanization, are also associated with aspirations. We recommend a "whole-family" approach, which acknowledges the influential role of parents, for policies and programs for rural youth and a stronger focus on gender aspects.

☆We are grateful for all the farm families participating in the study and their willingness to share their aspirations with us. The fieldwork was only possible thanks to the help of excellent and dedicated research assistance from a team around Fabiano Tembo from Kachele Research Solutions and the Indaba Agricultural Policy Research Institute (IAPRI). We are grateful for the financial support from the "Program of Accompanying Research for Agricultural Innovation" (PARI), which receives funds from the German Federal Ministry of Economic Cooperation and Development (BMZ). We would like to thank the excellent reviews and comments from the three anonymous reviewers and the editor.
Address correspondence to Thomas Daum, Division of Social and Institutional Change in Agricultural Development, Hans-Ruthenberg-Institute of Agricultural Science in the Tropics, University of Hohenheim, Wollgrasweg 43, 70599, Stuttgart, Germany. Email: thomas.daum@uni-hohenheim.de

Introduction

There is an increasing focus on the role of youth in rural transformation processes. Governments across Africa are promoting "youth in agriculture," arguing that agriculture is key to providing the much-needed jobs for the millions of youth[1] entering Africa's job markets (Bullock and Crane 2021; IFAD 2019; LaRue et al. 2021). It is expected that by 2035, 220 million youth will have entered the African labor market (von Braun and Kofol 2017). Creating meaningful jobs is seen by many as key not only to the youth's livelihoods and well-being but also to curbing rural outmigration and ensuring peace and security (von Braun and Kofol 2017). The success of promoting agriculture as a livelihood for rural youth hinges on the rural youth wanting to stay and farm. Consequently, an emerging body of literature explores their aspirations (Bullock and Crane 2021; Chamberlin, Ramos, and Abay 2021; LaRue et al. 2021; Mussa 2020; Rietveld, van der Burg, and Groot 2020; Ripoll et al. 2017; Sumberg et al. 2017).

This literature has focused on both the strengths and direction of aspirations and has explored potential factors associated with youth aspirations—such as land access (Bezu and Holden 2014), rural vibrancy (Chamberlin et al. 2021), or, more generally, the socioeconomic opportunity space (Bullock and Crane 2021; Leavy and Hossain 2014; Leavy and Smith 2010). Much less is known about the actors shaping youth livelihood aspirations, particularly the role of parents. Moreover, studies have frequently portrayed youth as a homogeneous group, neglecting gender aspects (Bullock and Crane 2021; Rietveld et al. 2020). Lastly, although studies have focused on whether farming is attractive, few studies have explored how attractive farming looks, for example, regarding the types of technologies used (Daum 2019). This paper addresses these neglected aspects by using a unique "whole family" data collection approach in rural Zambia. The approach is based on data collected from different household members to reflect that male and female youth can have different aspirations and that these aspirations are not formed in a vacuum outside of their families. Compared to previous studies, which often collected data only from a single parent (typically the household head), this approach allows the elucidation of differences in father's

[1]Youth are often defined on age grounds such as 18–35 years by the government of Zambia and 15–24 years by the United Nations (IFAD 2019). Many scholars have argued against the use of age-based definitions of "youth". Pereznieto et al. (2018:11) have defined youth as "a time of transition - from school to work, dependence on autonomy, and sexual maturity". In this study, we focus on adolescents from 12–18 years because this period is key for the formation of aspirations.

and mother's aspirations for their sons and daughters and to explore to which degree they shape youth aspirations (as further explained below).

Aspirations are forward-looking goals that people are committed to expending their energy, time, and resources toward (Sherwood 1989). As highlighted by LaRue et al. (2021), aspirations can be conceptualized as ambitions ("the strengths of a desire") (Leavy and Smith 2010), which will be referred to as *general aspirations* in this paper, or as what people would like to do (Leavy and Smith 2010), which will be termed as *livelihood aspirations*. Aspirations depend on personal characteristics, culture, traditions, norms, social networks, media, and the socioeconomic opportunity space (Bullock and Crane 2021; Gutman and Akerman 2008; Leavy and Hossain 2014; Leavy and Smith 2010; Wang, Hagedorn, and McLaughlin 2021). Several studies have questioned youth's farming aspirations due to push factors such as lacking access to land and finance (Bezu and Holden 2014; IFAD 2019) and the perception of agriculture as a back-breaking, unrewarding job for the uneducated (Sumberg et al. 2017). Other studies found that youth have more nuanced views (Daum 2019) and often want to farm in a mixed livelihood strategy (LaRue et al. 2021; Rietveld et al. 2020), or to keep their options open (Huijsmans et al. 2021).

Inter-generational aspects related to the formation of aspirations have been neglected despite emerging qualitative evidence suggesting that parents influence youth aspirations (Huijsmans et al. 2021; Muwi 2012; Wang et al. 2021). Studies from other fields show that parents often play a significant role in the formation of their children's career choices (Gutman and Akerman 2008; Paloş and Drobot 2010; Wiley, Bogg, and Ho 2005). Empirical studies in low- and middle-income countries suggest that parents often oppose a farming career for their children (Bullock and Crane 2021; Lungkang 2018; Verkaart, Mausch, and Harris 2018; Yeboah et al. 2017). However, such studies have often portrayed farming as a "take-it-or-leave-it" option (e.g., Verkaart et al. 2018), neglecting the potential role of part-time farming and often only collecting data from household heads. Moreover, little is known about whether the parent's views have any actual bearing on the aspirations of youth—or if other actors are more influential. In a study in Kenya, young people reported that the most influential actor category is their family. However, this category comprised parents, siblings, other family members, and relatives, and no gender differences were analyzed (LaRue et al. 2021). Inter-generational aspects may also influence youth aspirations through the experienced and expected future level of parents' support for certain livelihood decisions, for example, in the form of capital and land inheritance (IFAD 2019; Quisumbing et al. 2014; Rietveld et al. 2020).

Gender aspects are highly relevant in the context of youth in agriculture (Bullock and Crane 2021; Rietveld et al. 2020). Gender represents a socio-cultural construct defining the roles, responsibilities, constraints, and opportunities of men and women—and boys and girls. There is strong evidence of gender disparities in Africa at the societal and household level. For example, women are more likely to farm on smaller and lower quality land areas (Doss et al. 2018; Smale, Theriault, and Haider 2019) and have more limited access to agricultural inputs and services (Quisumbing et al. 2014). Across 42 countries, IFAD (2019) reports that young women are half as likely to own land and twice as likely to be out of school and work, often because of early marriage and child-rearing responsibilities. Seeing and experiencing these gender disparities can have a direct bearing on the aspirations of adolescent boys and girls. In addition, socially defined gender norms may shape aspirations toward farming (IFAD 2019). Lastly, adults may have different aspirations and provide different levels of support for boys and girls, thereby affecting their aspirations. Empirical studies suggest that male youth are more likely to aspire to a career in farming than their female peers (Elias et al. 2018; Rietveld et al. 2020). Gender disparities may also affect youth perceptions of a "perfect farm" (Daum 2019). Understanding gender differences regarding youth in agriculture is essential for well-grounded policy interventions promoting youth engagement in agriculture (Pereznieto et al. 2018).

This study followed a unique "whole family" approach for empirical data collection in Zambia to explore the knowledge gap on generational and gender phenomena and their influence on youth livelihood aspiration. As part of this multigenerational approach, we interviewed 348 household heads and spouses alongside sons and daughters from 87 rural households in the Eastern Province of Zambia. The sampled households were located in communities with different proximities to urban areas. Data were collected using a mixed-methods approach, combining a quantitative survey and qualitative tools such as focus group discussions, which helped to reveal aspects such as traditions and social norms. The paper addresses several objectives: (1) to examine the livelihood aspirations of boys and girls as well as the livelihood aspirations of parents for their sons and daughters, (2) to explore to which extent parent's general and livelihood aspirations—as well as other factors and actors—shape youth livelihood aspirations, and (3) to explore how the "perfect farm" looks for boys and girls.

Conceptual Framework

Figure 1 illustrates various factors that can influence the formation of and actual aspirations of youth, focusing on parents and youth characteristics. As noted above, aspirations can be conceptualized, focusing on how much (level) and what people want to achieve (direction) in terms of general and livelihood aspirations. In this framework, we do not categorize farming as a "take-it-or-leave-it" option, which is a shortcoming of many studies on youths in agriculture (LaRue et al. 2021) but instead distinguish between full-time, part-time, and no farming-based livelihoods. These aspects will be examined using econometric tools (see also Table 2). Additional socio-cultural and economic factors that may influence the formation and actual aspirations of the youth (e.g., culture, tradition, and norms) will be examined with the help of qualitative tools (see Section "Methods and Sampling").

Parents' characteristics include their farm and household characteristics such as farm size, assets, degree of mechanization, and income. These aspects can determine how young people experience farming (e.g., as lucrative or burdensome). Moreover, they may influence the youth's opportunities and challenges when deciding to pursue farming. Land, for example, is typically inherited. In many low- and

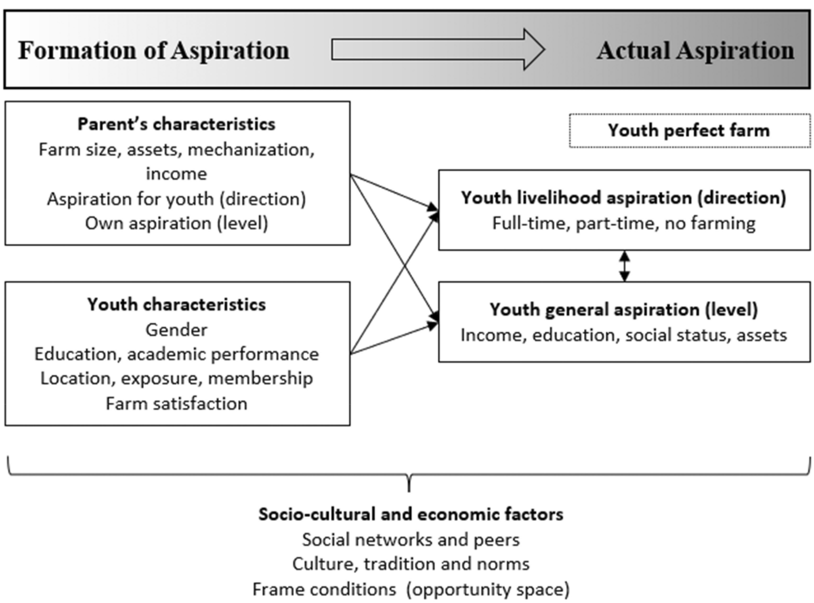

Figure 1. Conceptual Framework.

middle-income countries, the patriarchal system of inheritance favors sons, which may discourage adolescent girls from aspiring to farm (IFAD 2019; Quisumbing et al. 2014; Rietveld et al. 2020). Additional parental characteristics addressed are their own aspirations. Parents' aspiration levels can significantly impact their children's educational (Byun et al. 2012) and career aspirations (Jungen 2008). Their bearing may be explicit or implicit and can be exerted in the form of work ethics, family values, and gender stereotyping, among others (Jungen 2008). In addition, the parents' aspirational direction (livelihood aspiration) for their children may matter. Parents' aspirations can play out in the form of inspiration, consultation, trust, and support—or pressure and coercion (Paloş and Drobot 2010; Wiley et al. 2005). Parents' livelihood aspirations for their children may depend on the socio-economic condition of the family as well as broader economic and socio-cultural factors.

The youth characteristics studied in this paper include gender, education, academic performance, location, exposure, group membership, and farm satisfaction. The importance of gender has already been elaborated on. Education and academic performance are significant for aspiration development because of their role in expanding the opportunity space (Anyidoho et al. 2012; Bezu and Holden 2014). Studies suggest that young people with more education stand a better chance of securing jobs in the formal sector (Bessant 2018; Leavy and Hossain 2014). Moreover, young people may view farming as an unattractive occupation more for school drop-outs (Tadele and Gella 2012). Remoteness may support farm-based aspirations as youths in remote areas have fewer non-farming job opportunities, which may also be more burdensome and less rewarding, and because rural areas have been associated with close communal ties that make young people favor the status quo (Daum 2019; Leavy and Smith 2010). However, remoteness may be a discouraging element as remote areas often suffer from a lack of infrastructure and services such as good roads, markets, schools, and hospitals. Aspirations are also affected by what young people "know or can imagine" (Bajema, Miller, and Williams 2002:62) and qualitative studies have established a link between exposure to media, urban areas, and livelihood aspirations (Bernard et al. 2014; Leavy and Smith 2010). Social groups can serve as a platform for encouraging young people into agriculture—as such groups often provide teaching on innovative farming methods (FAO, IFAD, and WFP 2015)—or a platform where youth get exposure to alternative livelihoods. Lastly, farm satisfaction may shape youth aspirations as many rural children help on their parent's farms (Anyidoho et al. 2012; Huijsmans et al. 2021; Lungkang 2018). In many poor households, parents employ children as readily available cheap labor (Huijsmans et al. 2021). Farm satisfaction is interrelated with household characteristics such as the use of hired labor and mechanization (Huijsmans et al. 2021) and may differ depending on the typical tasks of boys and girls.

Study Site, Sampling, and Methods

Study Site: History, Agriculture, Land, Gender, and Youth

Agriculture is key to the livelihoods of more than 90% of the rural population in Africa (Davis, Di Giuseppe, and Zezza 2017) and 85% of the population in Zambia (Ali et al. 2019). Zambian agriculture is characterized by a large-scale farming sector (i.e., along the "line-of-rail" in the Southern, Lusaka, and Central Provinces) and a small-scale farming sector (Üllenberg et al. 2017). This study focuses on the Eastern Province of Zambia, which is one of the largest smallholder agricultural regions where agriculture is the primary source of livelihood for most of the rural population. The Eastern Province represents the type of smallholder mixed farming systems that are common across various African countries. The Eastern Province is also the site of changing opportunities and challenges related to farming and of contested agricultural development pathways, both historically and contemporary, as further outlined below.

Land cultivation in the Eastern Province is mainly under customary tenure, with vacant land allocated by chiefs and village heads (Amberntsson 2020; Gumbo et al. 2016). 54% of the land (Sitko and Chamberlin 2016) and 88% of agricultural parcels are under customary law (Ali et al. 2019). Few households possess formal documents, and 67% of the farm households perceive their lands to have insecure tenure (Ali et al. 2019). 16% of the parcels are associated with a formal right to bequest but land can be transferred within families as part of the customary law (Ali et al. 2019). Under customary law, the land is typically inherited by male relatives or children, as bequeathing land outside of the male bloodline is considered taboo and against the ancestors (Chapoto, Jayne, and Mason 2011). This patrilineal inheritance system means women suffer from high levels of tenure insecurity (Ali et al. 2019; Chapoto et al. 2011). Males control 79% of the parcels (Ali et al. 2019). Women typically have access to land through male relatives (i.e., husbands or fathers) but with little control and security (Chapoto et al. 2011; Gumbo et al. 2016). In some areas of the Eastern Province, matrilineal systems are common, but they were not part of the randomly sampled study sites. While Zambia and its Eastern Province are considered sparsely populated and land-abundant in general, there are rising land scarcities in areas close to large towns and roads where market access is favorable, leading to land commodification and alienation (Sitko and Chamberlin 2016).

In the Eastern Province, 57% of the farmers cultivate less than 2 hectares of land, mostly growing maize, groundnuts, sunflower, and soybeans, which can all be grown for subsistence and as cash crops, with some additional cotton cultivation (Amberntsson 2020). Yields are modest, with an average maize yield of 1.7 tons (IAPRI 2019). Many farm households have small gardens, growing fruits, and vegetables (Amberntsson 2020). 83% of all households own chicken—10 on average—and 50% own cattle—on average 6 (IAPRI 2019). A significant share has no animals at all (Amberntsson 2020). Less than 1% engage in aquaculture (IAPRI 2019). On average, farmers sell 33% of their agricultural produce, however, around 17% of all households are pure subsistence farmers (IAPRI 2019). On average, 65% of the income of rural households comes from agricultural production (mainly from the sales of crops but also the sales of fruits and vegetables and livestock products) (IAPRI 2019). Pelekamoyo and Umar (2019) show that household decisions, including income allocation, are increasingly done jointly; however, men often still have more decision-making power over aspects that were traditionally a "male domain." For example, while women have some control over small livestock, they typically have limited control over large livestock such as cattle (Lubungu and Birner 2021).

Most households depend fully on manual labor except for land preparation, during which 67% use animal draught power and 2% use tractors on at least parts of their land (IAPRI 2019). It is often believed that most labor in agriculture comes from women. However, recent studies suggest an equal contribution of men and women to agriculture, with some differences depending on the tasks (e.g., males contributing slightly more during land preparation, and females contributing slightly more during weeding and processing) and the degree of mechanization (e.g., indicating a declining contribution by women with increased mechanization) (Adu-Baffour, Daum, and Birner 2019; Daum and Birner 2021;Daum, Capezzone, and Birner 2021; Pelekamoyo and Umar 2019). Most of the household chores and care activities are conducted by females (Daum, Capezzone, and Birner 2021; Pelekamoyo and Umar 2019). Both boys and girls spent a significant amount of their time and energy on the fields (Daum and Birner 2021; Daum, Capezzone, and Birner 2021). As noted by Kanduza (1987) and still valid today (see Section "Overview of Farm Households and Gender Norms and Roles"), the socialization of children to agricultural labor in many households begins from around the age of 10. The children start with "light" activities such as planting, bird scaring, and looking after small livestock but later engage in more arduous activities such as land preparation, weeding, harvesting, and processing, or herding cattle in the case of boys (see also Daum, Capezzone, and Birner 2021).

Farmers increasingly use external inputs from private markets, processing companies, or the Farm Input Support Programme (IAPRI 2019). For maize, an important food and cash crop, 40% of the farmers use improved seeds, 73% use fertilizer, and 13% use herbicides (IAPRI 2019). According to Amberntsson (2020), many farm households suffer from depleted and degraded soils, a lack of chemical fertilizers, infrastructure and market deficiencies, and exploitation. Various donor- and government-led development projects focus on "farming as a business, assuming that farmers'" problems result from poor attitudes and capabilities (Amberntsson 2020), and there are various contract farming schemes by large agro-processing companies.

Amberntsson (2020) shows that rural households have diversified their income sources during the last decades but still identify themselves mainly as farmers, with off-farm incomes reinvested into farming activities. Amberntsson (2020) also found this to be the case for younger farm households. Statistics show that 60% of the households have a business; primarily retailing/vending, charcoal production, and local beer brewing (IAPRI 2019). 78% of the households earn less than 1.25 USD per day (IAPRI 2016). Zambia ranks 113 of 117 countries according to the Global Hunger Index (von Grebmer et al. 2019), with food insecurity being widespread during the "hunger months" (Amberntsson 2020; Daum et al. 2021).

Amberntsson (2020), Kanduza (1987, 1992), Vail (1977), and Üllenberg et al. (2017) provide comprehensive accounts of the historical factors shaping rural communities in Zambia. Before colonization, the Zambian territory was settled by different ethnic groups, primarily practicing shifting cultivation and hunting and gathering (Üllenberg et al. 2017). The British colonizers focused on copper mining (Kanduza 1987, 1992) whereas rural areas such as the Eastern Province were utilized as "labor reservoirs" of working-age male adults (Amberntsson 2020; Kanduza 1992; Üllenberg et al. 2017; Vail 1977). The colonizers organized agricultural production around large-scale, European-owned farms along the aforementioned "line-of-rail" whereas African rural households were discriminated (Amberntsson 2020; Gumbo et al. 2016; Kanduza 1987; Üllenberg et al. 2017; Vail 1977). This changed during the 1950s when the most "progressive" farmers were encouraged to produce for the market and received support as part of "Peasant Farming Schemes" and "Improved Farming Schemes" (Amberntsson 2020; Kanduza 1987).

After independence, during a period of high copper prices, the government depended on copper mining and tried to diversify, promoting industrialization and large-scale farming (Üllenberg et al. 2017). Following a decline in copper prices, the government focused on smallholder agriculture in the early 1980s (Kanduza 1987). Farmers were supported with subsidized inputs, credit, fixed prices, infrastructure development, and extension services, leading to two decades of development and progress (Amberntsson 2020; Gumbo et al. 2016). However, the heavy state support could not be sustained and had to be cut back partially as part of the structural adjustment reforms in the 1990s (Amberntsson 2020; Gumbo et al. 2016). Since the 2000s, the idea of "farming as a business" gained popularity, and private sector companies began to play an increasing role in development efforts, for example, as part of out-grower schemes (Amberntsson 2020; Gumbo et al. 2016). In addition, subsidized inputs again starteplay a significant role in form of the Farmer Input Support Programme (Gumbo et al. 2016; Mason, Wineman, and Tembo 2020). Between the mid-2000s and mid-2010s, a boom in copper mining triggered rapid urban economic growth, enabled large infrastructure projects, and created many employment opportunities across Zambia, leading to average annual gross domestic product growth (GDP) rates of around 7% and a drop of youth unemployment from 28% to 16% (World Bank 2021a, 2021b). Between 2015 and 2019 (the year of our data collection) falling copper prices, energy shortages, and droughts, among others, led to stalling GDP growth (around 3%) and rising youth unemployment, reaching 23% in 2019 (World Bank 2021a, 2021b). Since the 2010s, farmers have increasingly suffered from climate change, affecting the yields and riskiness of farming (Mulenga, Ngoma, and Tembo 2015).

Zambia is characterized by a very young population, with 54% under 18 years old (UNICEF 2021). According to UNICEF (2021), 59% of all children across Zambia live in poverty or extreme poverty. The primary and secondary school enrolment rate rose significantly over the last decades for both boys and girls, standing at 88% and 43%, respectively (with a gender parity rate of 1 at the primary level and 0.9 at the secondary level) (UNICEF 2021). According to UNICEF (2021), around one-third of girls and two-fifths of boys experience violence (i.e., from parents and relatives) before the age of 18. Zambia has high rates of teen pregnancy and child marriage, with around one-third of all women marrying before 18 (UNICEF 2021).

Methods and Sampling

This study used both quantitative and qualitative methods. To select respondents for the quantitative questionnaire, we used a multistage sampling technique. First, we randomly sampled three districts (Lundazi, Mambwe, and Vubwi) from the nine districts of the Eastern Province (see Table 1 and Figure 2). Second, we randomly selected agricultural camps—administrative units used by the extension service—in each district based on "remoteness." For this, all camps were categorized as urban, peri-urban, and rural, based on characteristics such as distance to markets, roads, hospitals, banks, and district centers, and one camp representing each of these categories was randomly selected: Mapala, Masumba, and Sekani, representing urban, peri-urban, and very remote study areas, respectively. In each of the camps, comprehensive lists of farm households were obtained from the local agricultural extension officers. The register was used to randomly select 30 households in each camp,

Table 1. Distribution of the Sample Size

District	Households	Men	Women	Boys	Girls	Total
Lundazi	27	27	27	27	27	108
Mambwe	30	30	30	30	30	120
Vubwi	30	30	30	30	30	120
Total	87	87	87	87	87	348

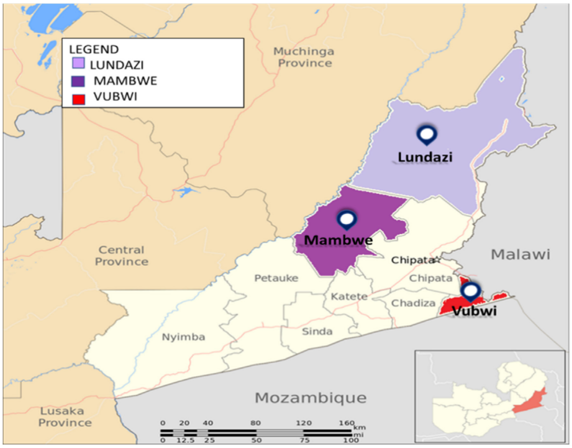

Figure 2. Map of the Study Sites. [Colour figure can be viewed at wileyonlinelibrary.com]

Table 2. Description of the Variables Used in Regression Analysis

Variable Name	Variable Description
Outcome variables	
Full-time farming	Youth envision becoming full-time farmers (as compared to non-farming)
Part-time farming	Youth envision becoming part-time farmers (as compared to non-farming)
Explanatory variables	
Youth characteristics	
Education	Highest grade completed
Academic performance	Self-estimated rank in class during the last term in school
Farm satisfaction	Satisfaction on a scale from 0–10, where 10 is the highest level
Exposure	Youth exposure index (see Section “Youth exposure index”)
General aspiration level	Youth aspiration index (see Section “Aspiration index”)
Land inheritance	Estimated share of farmland in ha to be inherited from parents
Location	Degree of remoteness (urban, peri-urban, and rural)
Desired location	Desired place of future residence (rural, urban, foreign)
Group membership	Membership in a social group (1 = yes; 0 = no)
Parents characteristics	
Household farm size	Farmland owned in ha
Assets	Number of assets owned from list of ten assets: (i) tractor or two-wheeled tractor, (ii) set of animals and equipment for animal draught power, (iii) car/van/truck, (iv) trailer, (v) motorcycle, (vi) bicycle, (vii) television, (viii) radio, (ix) mobile phone, and (x) refrigerator
Farm income	Share of farm income relative to total household income
Crop income	Share of crop income relative to total farm income
Livelihood aspiration for sons/girls	Full-time farming, part-time farming, no farming
General aspiration level	Aspiration index (see the index, Section “Aspiration index”)
Power Source	Main source of farm power: own manual labor, animal traction, or hired labor (none of the selected households accessed tractors)

excluding polygamous households to avoid complex and confounding family structures. Another condition was that the households must have both daughters and sons aged 12–18 years. In each of the selected households, the household heads, spouses, and boys and girls were interviewed with the help of local enumerators, ensuring that only female assistants conducted interviews with female respondents. All household members were interviewed separately to ensure the free expression of their opinions. Overall, 87 households agreed to participate in the study, leading to a total of 348 respondents (see Table 1). The questionnaires were administered during June and August 2019. Informed consent was obtained from all study participants. For respondents below the age of 18, consent was given by themselves as well as their parents.

The questionnaire was complemented with qualitative methods to elicit information on issues that cannot easily be captured using quantitative methods. In each of the selected agricultural camps, five FGDs were conducted: (1) with household heads (2) with spouses, (3) with boys (12–18 years), (4) with girls (12–18 years), and (5) with young men and women aged 16–25 years who were just starting to build their own families, to allow for a broad spectrum of views on youth in agriculture. Each FGD consisted of 6–9 randomly selected participants with different social-economic backgrounds regarding wealth, status, and education. A total of 15 FGDs were conducted.

Data Collection and Analysis

Aspiration index. We explore both the direction of aspirations (livelihood aspiration), in particular, whether youth aspire to be full-time farmers, part-time farmers, or no farmer, and the strengths of aspirations (general aspiration). For the latter, an aspiration index was used that rates ambitions regarding income, education, social status, and assets (Bernard and Taffesse 2014; LaRue et al. 2021). Male and female youth were asked questions (1) and (3) to determine their aspirations and adults were asked questions (2) and (3) to ascertain their aspirations while growing up:

1. On a scale of 0–10, which level of income (education, social status, and assets) would you like to achieve 10 years from now?
2. On a scale of 0–10, which level of income (education, social status, and assets) did you want to achieve when you were 20 years old?
3. Which of the dimensions are most important for you? Here are 20 beans. Please distribute all 20 beans across the four dimensions according to their importance.

We then used the standardized method to compile an aggregated aspiration index as developed by Bernard and Taffesse (2014). The index was constructed by normalizing the desired level of the various dimensions by subtracting the average level for individuals in the whole sample and dividing the difference by the individual's standard deviation across the entire sample. The outcome was multiplied by the weight that individuals assigned to the various aspects of aspirations (see question 3). The aggregated aspiration index was produced by summing up the weighted average of the four normalized outcomes. The aspiration index is written as:

$$A_i = \sum m \left(\frac{a_i^m - \mu^m}{\sigma^m} \right) * W_i^m$$

where

a_i^m: individual i's response to questions one (youth) and two (adult) above concerning dimension m (income, education, social status, or assets).

μ^m: sample mean for responses to questions (1) and (2)—the youth and adult's aspired outcomes for dimension m.

σ^m: standard deviation for responses to questions (1) and (2)—the youth and adult's aspired outcomes for dimension m.

W_i^m: weight that individual i assigned to the corresponding dimension when answering the question (3).

Youth exposure index. To assess youth exposure beyond the social network, we constructed an index based on answers to four questions:

1. *"How often do you listen to the radio?"*
2. *"How often do you use the Internet?"*
3. *"How often do you watch television?"*
4. *"In the last year, how often have you gone to the province capital?"*

Every question requires a response and a score was assigned: (a) never (1), at least once a year (2), at least once a month (3), at least once a week (4), every day (5). Hence, the maximum overall score was 20 points (4 questions * 5 points), and the minimum score was 4 points (4 questions * 1 point). Higher scores reflect higher levels of exposure.

Social networks. To study the actors influencing the youth, we developed a simple participatory visual tool. Respondents received 50 beans and a large, printed grid with squares representing different actors such as father, mother, brother, sister, relatives, friends, media, church, youth associations, school, NGOs, and extension (agricultural ministry). Respondents were allowed to add additional squares representing other actors. Respondents distributed the 50 beans across the squares according to how much the different actors affect their aspirations. The values assigned can be expressed as percentages. For example, assuming a respondent placed 20 beans on the father, 15 on the mother, and 15 on the brother, this can be expressed as a 40% importance to the father, 30% to the mother, and 30% to the brother. Respondents were also asked about the direction of influence (if the respective actors encourage or discourage them to pursue farming).

"Perfect farm". A similar approach was adopted to understand the respondent's "perfect farm." Here, participants were asked to distribute 50 beans across various farm features such as access to market, tractor, animal draught power, ICT, environmental friendliness, access to extension service, input use, agro-forestry, social aspects, livestock, horticulture, access to water, fish, and land that we identified based on preliminary focus group discussions. Again, the respondents had the opportunity to choose additional items besides the listed items.

Data analysis. The study used multinomial logistic regressions to explore factors correlated with the livelihood aspirations of boys and girls in agriculture (full-time, part-time, or non-farming). Multinomial econometric regressions were used because of their ability to assess the factors associated with the probability of belonging to a specific categorical group (Peng et al. 2002; Peng and Nichols 2003). Table 2 provides an overview of all variables used in the multinomial logistic regressions.

Results

Overview of Farm Households and Gender Norms and Roles

Previously, farming was practiced for subsistence reasons and prestige, but it is now increasingly seen as a business according to the qualitative insights from the 15 FGDs from the selected rural communities (see Section "Methods and Sampling"). Farming is the primary source of livelihood for most rural households, providing both food and income. This fact mirrors the insights from the quantitative questionnaire. Table 3 provides an overview of the sample characteristics of the 87 households, showing answers from heads and spouses, as well as the arithmetic means. (In the subsequent analysis, we will use the arithmetic means between the two answers). Households derive 86% of their household income from farming, of which 73% is from crop farming. 14% of household income comes from off-farm work. Confirming the evidence from the literature, the qualitative insights from the 15 FGDs suggest that women have less control over productive resources. While women typically have access to land (through their husbands or male relatives), they have little control over land, and when land is given to women, it is often small and of low quality. According to the qualitative insights, despite their large labor contribution to agriculture, women have limited power over farm decision-making and income, as the following quotes illustrate:

> The problem is that my husband does not involve me in decision-making on crops to plant and how to use the money from the sales of the farm produce. (Female FGD participant, 22 years, Lundazi)

> The problem for me is my husband. He drinks a lot and most painfully uses the money from the sale of our farm produce to pursue other women. (Female FGD participant, 25 years, Lundazi)

Table 3. Descriptive Statistics of Households

Variable	Head (*N* = 87) Mean	Std. Dev	Spouse (*N* = 87) Mean	Std. Dev	Average (*N* = 87) Mean	Std. Dev	Mann–Whitney *p*-Value
Individual-level							
Age (years)	49	7.60	44	7.24	47	7.94	.000
Education (grade)	6th	2.88	4th	2.64	5th	3.01	.000
Household-level							
Household size	–	–	–	–	9	1.76	–
Farm size (ha)	5	3.9	5	3.8	5	3.83	.866
Income shares (%) …							
from farm	87	22.7	85	25.3	86	24.03	.766
from crops	72	26.7	74	24.1	73	25.41	.723
from livestock	16	18.7	15	18.6	15	18.57	.784
from off-farm	10	19.7	13	24.4	12	22.15	.853
from wage/salary	2	8.8	1	7.7	1	8.26	.416
Assets (#)	–	–	–	–	3	1.64	–
General aspiration index	8.71	23.88	4.29	25.59	6.50	24.78	.448

Note: Arithmetic means are rounded.

> The man always handles the money garnered from the sales of our farm produce. He often uses that money to chase other women (…). As a result, it becomes impossible to make progress in the farming business. (Female FGD participant, 20 years, Lundazi)

However, the focus group discussion also suggests a gradual change in gender norms and roles. Partly because of advocacy campaigns by development projects, women have increased access to inputs such as fertilizer and seeds and growing decision-making power over some aspects of farming such as small livestock production and kitchen gardens. Moreover, joint decision-making is also on the rise in some households. Labor dynamics are changing too. For example, groundnuts were considered a female crop due to their low economic value in the past but nowadays groundnuts (as well as soybean) are considered cash crops involving the labor of both men and women.

The qualitative insights from the focus group discussion also suggest changing gender norms and roles related to boys and girls. For example, the access of girls to education has improved rapidly and many households stated they do not prefer boys over girls when making investments into educati. Nevertheless, in some households, parents are still more reluctant to support girls' education. At the same time, opportunities

Table 4. Descriptive Statistics of Adolescent Boys and Girls

	Boys (*N* = 87)		Girls (*N* = 87)		Mann–Whitney
Variable	Mean	Std. Dev	Mean	Std. Dev	*p*-Value
Age (years)	16	2.07	15	2.01	.001
Education (grade)	6	2.37	6	1.99	.107
School satisfaction[a]	9	2.81	9	2.60	.484
Academic performance (rank)	9th	0.13	13th	0.17	.359
Farmland owned (ha)	0.13	0.36	0.06	0.32	.030
Share land to be inherited (%)	22	27.63	3	8.37	.000
Weekly income ($)	6	1.75	2	3.00	.000
Exposure index	8.44	2.44	7.41	2.83	.007
General aspiration index	15.44	20.23	10.27	24.20	.352

[a]School satisfaction based on scales from 0–10 where 10 was the highest level of satisfaction.

and expectations for girls to not only marry but also join the formal labor market have risen. Table 4 shows the sample characteristics of the selected boys and girls from the 87 households you were interviewed as part of the quantitative data collection (see Section "Conceptual Framework"). Boys owned double the farmland area (0.13 ha) compared to girls (0.06 ha) and were estimated to "inherit" a higher percentage of the parent's farmland than girls (22% as compared to 3%). These statistics and the qualitative interviews confirm the literature on the patrilineal system of inheritance (Section "Study Site: History, Agriculture, Land, Gender, and Youth"). When a male youth reaches maturity, he can approach the family for a piece of land for farming. When the family lacks enough farmland, the household head can approach the village head for land. Girls are rarely considered in the distribution of arable land for farming purposes based on the belief that the female youth can access land from the future husband, as reflected in the following quotes:

> If I must share land between my children, I will share it unequally in favor of the male child because my female child's husband can easily gain control of the land when she gets married. (Female adult FGD participant, Lundazi)

> The male children are given more preference than females [because according to tradition they] retain the family name whereas the female child will be bequeathed to another family when she gets married. (Female FGD participant, 15 years, Mambwe)

However, the female youth can always return to the family in the event of divorce or death of the husband, wherein the male head of the household is expected to allocate her a piece of land for farming (access rights).

The qualitative evidence suggests that boys and girls participate actively in farming activities, with few gender differences between the different tasks, as the following quote suggests:

> There is no difference in the agricultural tasks given to boys and girls because doing so will encourage laziness, especially with girls. (Female adult FGD participant, Mambwe)

Table 5 provides insights into the self-reported time-use of boys and girls during an average week in the farm season (reflecting the respective farm season). In terms of farm work, boys and girls have similar workloads (days/week) except during the land preparation season when boys work five days per week and girls two days per week. The complementary qualitative evidence suggests that boys and girls work a similar amount of hours as well. Both boys and girls do domestic chores, but girls spend significantly more days per week and hours per day on such activities, including cooking and taking care of children/elderly. On average, boys reported being more satisfied with farming activities than girls, while girls reported higher satisfaction with domestic chores; however, social norms may prime this perception.

Boys and girls learn about farming from early childhood. In the first years, the children mainly observe their parents, but they soon start to help on the farm, beginning with "light" work such as bird scaring and planting. In their teenage years, boys (and some girls) are given a small portion of land as well as seeds and fertilizers from their parents to practice how to farm:

> Knowledge on farming-related activities is passed from parents to children from a tender age. (…) Over time, as they mature, we give the children (especially the boys) a small portion of land to cultivate while providing necessary guidance and assistance (…) to learn but also equip them for a future farm livelihood in the absence of a white-collar job. (Male adult FGD participant, Lundazi)

> Engaging the children in household farming enables us to prepare them, especially the males, for future survival as the family's breadwinner. (Female adult FGD participant, Lundazi)

Table 5. Time-Use Activities in Days Per Week of Boys and Girls

	Boys (N = 87)		Girls (N = 87)		Mann–Whitney
	Mean	Std. Dev	Mean	Std. Dev	p-Value
Agricultural activities					
Land preparation (d/wk)	5	2.12	2	2.54	.000
Planting (d/wk)	3	2.04	4	1.95	.134
Weeding (d/wk)	4	2.04	4	2.10	.304
Harvesting (d/wk)	3	2.03	3	2.09	.380
Processing (d/wk)	3	2.04	4	2.17	.516
Farm satisfaction[a]	8	2.04	6	2.49	.000
Domestic activities					
Cooking (d/wk)	1	1.27	4	2.26	.000
Cleaning (d/wk)	1	1.62	5	1.87	.000
Washing (d/wk)	2	1.30	3	1.89	.000
Child/old/sick care (d/wk)	0	1.10	3	2.90	.000
Fetching water (d/wk)	3	2.45	6	1.71	.000
Collecting firewood (d/wk)	1	0.93	1	1.23	.207
Overall domestic chores (h/d)	2	1.08	3	1.15	.000
Domestic chores satisfaction[a]	5	2.85	8	1.85	.000

Note: Time-use for agricultural activities refers to the respective farming step (e.g., time spent on harvesting refers to a typical week during harvesting).

[a]Satisfaction was asked using scales from 0–10, where 10 was the highest level of satisfaction. Arithmetic means are rounded.

Boys and girls also receive farming-related training in school as part of so-called "production unit" lessons where they cultivate a small portion of land under the supervision of their teachers, to provide a "hands-on" learning experience on different farming operations.

What are the Aspirations of Adolescent Boys and Girls?

As a first step, we asked the youth about their livelihood aspirations in an open question to avoid framing effects. Figure 3 indicates that more young men (38%) desire to farm than young women (20%). Young females (36%) considered becoming a teacher to be a more desirable career than young males (29%). Girls more often envisioned themselves becoming nurses (24%), while boys (20%) saw themselves becoming doctors more often. For many girls, aspirations were not only occupational

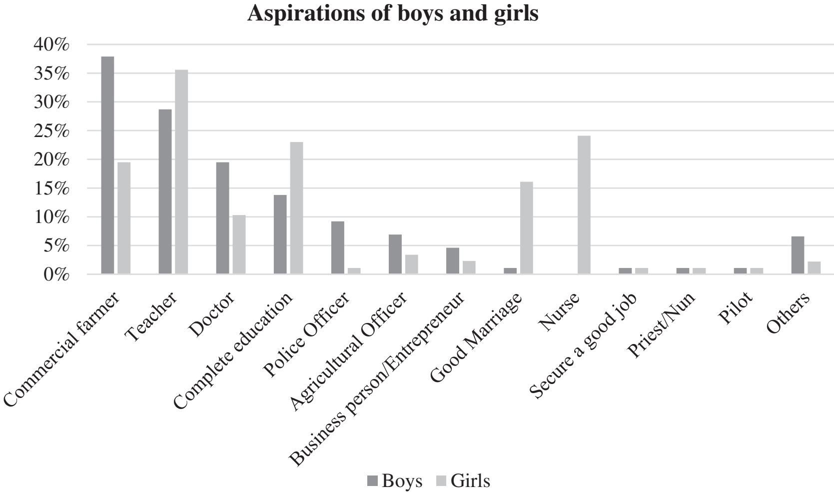

Figure 3. Livelihood Aspirations of Adolescent Boys and Girls.

Table 6. Distribution of the Livelihood Aspirations of Youth in Agriculture

	Boys (***N*** = 87)	Girls (***N*** = 87)
Full-time farming	40%	26%
Part-time farming	45%	35%
No farming	15%	39%

but also educational and marital. 16% of the girls aspired to a "good marriage" and 23% to "complete education."

When asked specifically about farming and allowing respondents to choose between full-time farming, part-time farming, and no farming, more boys (40%) desired full-time farming careers than girls (26%) (see Table 6). In addition, more boys (45%) aspired to participate in farming on a part-time basis than girls (35%). In contrast, more girls (39%) were not aspiring to engage in agricultural livelihoods as compared with boys (15%).

Among the reasons for the girl's disinterest in agricultural livelihood was the lack of access to land and the lower level of tenure security:

> Unfortunately, land allocation tends to favor males due to traditions, making it difficult for us females who desire to engage in farming on a large scale. (Female FGD participant, 16 years, Vubwi)

> A discouraging factor for me in farming is the lack of access to land. It can be very frustrating [to] not have enough land to cultivate because traditions prefer males. (Female FGD participant, 22 years, Lundazi)

In contrast, male youth typically have no problem obtaining land:

> It is not difficult to access a piece of land for farming if you grow up within the community. When I was about to get married and could not get sufficient family land, I approached the village head who then took me to an uncultivated bushy area to clear for farming. (Male FGD participant, 25 years, Vubwi)

Other reasons for the girl's disinterest in agricultural livelihoods related to the physical demand of farming as well as the increasing riskiness of farming due to climate change, as the following quotes show:

> Most of the farm operations are done manually and such old-fashioned practices can be very stressful for females who still have to do some household chores after returning from the farm. (Female FGD participant, 15 years, Lundazi)

> In this age where the weather is so unpredictable, you can incur huge loss from farm investment due to droughts or floods. This is unlike when you get a government job such as teacher or nurse or in the private sector where you're guaranteed regular salary. (Female FGD participant, 15 years, Mambwe)

Female youth who considered part-time farming explained their rationale for income diversification and raising funds to realize their aspiration of securing jobs in the formal sector:

> I will embrace farming as a part-time occupation to diversify my family income by combining income that comes through a regular salary as a policewoman to the income from farming. (Female FGD participant, 15 years, Vubwi)

> I will consider taking up farming temporarily to raise money and food required to realise my goal of becoming a teacher. (Female FGD participant, 12 years, Vubwi)

What are the Parents' Livelihood Aspirations for their Adolescent Sons and Daughters?

Figure 4 shows parents' livelihood aspirations for children when asked an open question to avoid framing effects. Although 37% of the fathers desired their sons to pursue farming (37%), only 12% desired their daughters to become farmers. Most (43%) aspired girls to have a "good marriage." Given the patrilineal inheritance system, this may reflect parents' wish to ensure access to land for their daughters. Similar to fathers, mothers more often aspired for their sons (37%) to become farmers than their daughters (14%) and more often aspired to a "good marriage" for their female children (53%).

Table 7 shows the results from the quantitative questionnaire when asking parents to choose between full-time farming, part-time farming, and no farming for their children. Both fathers and mothers were more likely to favor full-time or part-time farming for boys—and part-time or no farming for girls. One reason for the higher likelihood of favoring farming for boys relates to the land inheritance system:

> It is easier for boys to venture into farming because the system of land inheritance prioritizes males, particularly those that are married or of marriageable age. The belief is that men, as the family's breadwinners, require economic support for their responsibilities, unlike females whose needs are mainly catered for by the husband.
> (Male adult FGD participant, Vubwi)

The FGDs revealed that both fathers and mothers believed farming to be a lucrative business that can help male children to meet their

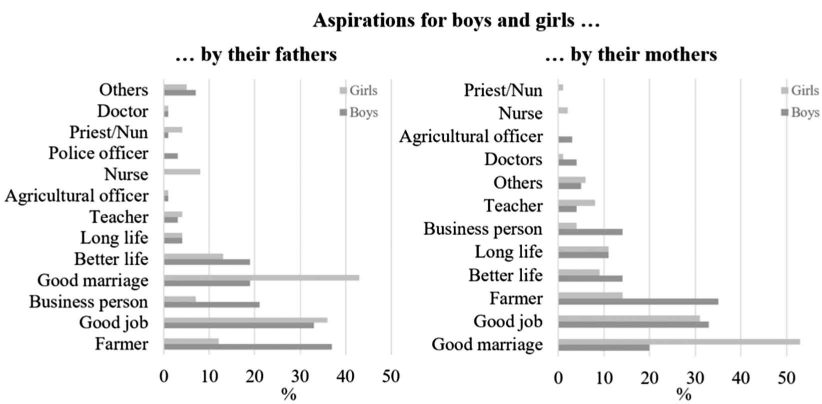

Figure 4. Parents' Livelihood Aspirations for their Sons and Daughters. Multiple responses were possible.

Table 7. Parent's Aspirations in Agriculture for Sons and Daughters

	Fathers' Livelihood Aspirations (N = 87)	Mothers' Livelihood Aspirations (N = 87)
… *for boys*		
Full-time farming	51%	52%
Part-time farming	38%	33%
No farming	11%	15%
… *for girls*		
Full-time farming	31%	30%
Part-time farming	48%	46%
No farming	21%	24%

responsibilities as future family heads in the absence of white-collar jobs. Some mothers also desired (part-time) farming for their male children to preserve the family farm tradition:

> Farming is a very lucrative business with great leverage potential for young boys nowadays in the absence of white-collar jobs, allowing them to meet family obligations. (Male adult FGD participant, Lundazi)

> I want my son to engage in part-time farming just to preserve the farm family tradition. (Female adult FGD participant, Vubwi)

Mothers' preferences for a non-farm career for female youth were associated with an understanding that young boys have better access to productive resources such as arable land and the labor burden associated with agriculture:

> Although agriculture is very much profitable nowadays when cultivated on a commercial scale, farming is still very challenging for females because they are not as strong as the males in withstanding the stress associated with farming manually. (Female adult FGD participant, Mambwe)

The motivation for some mothers who desire part-time farming for their daughters is to diversify income and ensure food security:

> I prefer my daughter to practice farming on a part-time basis to diversify family income, ensure food for home consumption, and to ensure she has something to fall back on in the event of job loss. (Female adult FGD participant, Vubwi)

Which Factors and Actors are Associated with the Livelihood Aspirations of Boys and Girls?

Table 8 shows different youth-related factors that may shape boys' aspirations toward farming based on the quantitative data from the household survey, which was analyzed with the help of multinomial logistic regression (see Sections "Methods and Sampling" and "Data analysis"). Pearson correlation analysis, tolerance values, and variance inflation factors suggest no multicollinearity problems. Boys' probability of envisioning a full-time or part-time livelihood in agriculture relative to a non-farm livelihood was significantly and positively correlated with farm satisfaction. Indicating a rural area as the desired place of residence (compared with residing in a foreign land) was positively associated with the livelihood aspiration to become full-time farmers. Factors such as education and academic performance, youth exposure, group membership, general aspiration levels, location, and the estimated share of land inherited were not correlated with the livelihood aspirations.

Table 8. Youth-Related Factors Associated with Boys' Aspirations toward Agriculture

	Full-Time Farming		Part-Time Farming	
	Estimate (S.E)	Exp (B)	Estimate (S.E)	Exp (B)
Education	0.102 (0.262)	1.107	0.188 (0.239)	1.207
Academic performance	−2.496 (3.693)	0.082	−1.244 (3.215)	0.288
Farm satisfaction	0.768*** (0.296)	2.156	0.539*** (0.203)	1.714
Exposure	0.074 (0.230)	1.077	0.170 (0.188)	1.186
Aspiration level	−0.060 (0.039)	0.942	−0.016 (0.039)	0.985
Land inheritance	0.005 (0.023)	1.005	0.009 (0.021)	1.009
Location (ref: peri-urban)				
Close to urban	−0.029 (1.480)	0.972	−2.021 (1.298)	0.133
Very remote area	0.691 (1.202)	1.996	−0.450 (1.031)	0.638
Desired location (ref: foreign)				
Rural	19.178*** (1.472)	2,133	2.208 (2.192)	9.099
Urban	17.170 (0.000)	2,862	1.574 (1.650)	4.826
Group membership	−1.598 (1.063)	0.202	−0.421 (0.911)	0.656
Constant	−22.422*** (3.443)	–	−5.770* (3.356)	–

Note: Cox & Snell R^2 (=0.501) and Nagelkerke R^2 (=0.577) suggest that a high share of the variations in the dependent variable are explained by the independent variables. Significance is indicated with ***p<.01, **p<.05, *p<.1. The exponentiated coefficients (Exp (B)) indicate to which extent a unit change in the explanatory variable changes the odds of falling into the group full-time (or part-time farming) while keeping all other variables constant.

The quantitative data from the household survey also suggests that similar to boys, girls' probability of envisioning a livelihood in agriculture compared to a non-farm livelihood was related to farm satisfaction (see Table 9). Girls' general aspiration level was negatively correlated with aspiring to a full-time farming livelihood. Girls with high general aspirations are likely to want to leave full-time farming behind. Compared to girls residing in peri-urban places, girls residing in more remote places were more likely to envision full-time farming, and girls who wish to reside in urban areas were less likely to aspire to full-time farming. Interestingly, both education and academic performance are correlated positively with the wish to pursue part-time farming (but not full-time farming); hence, the common perception that farming is a last resort for the uneducated is not always valid once mixed livelihoods are considered (where farming is combined with other income streams).

The analysis of the quantitative data from the household survey also shows how parents' own aspirations and farm characteristics correlate with boys' livelihood aspirations (Table 10). Boys from households using

Table 9. Youth-Related Factors Associated with Girls' Aspirations toward Agriculture

	Full-Time Farming		Part-Time Farming	
	Estimate (S.E)	Exp (B)	Estimate (S.E)	Exp (B)
Education	−0.216 (0.496)	0.806	0.402* (0.229)	1.495
Academic performance	−14.279 (9.050)	0.001	4.920* (2.746)	137.058
Farm satisfaction	1.439*** (0.521)	4.214	0.456***(0.155)	1.577
Exposure	−0.314 (0.299)	0.731	−0.151 (0.161)	0.860
Aspiration level	−0.099*** (0.038)	0.906	−0.031 (0.020)	0.969
Land inheritance	−0.014 (0.074)	0.986	−0.007 (0.051)	0.993
Location (ref: peri-urban)				
Close to urban	0.972 (2.017)	2.644	−1.479* (0.887)	0.228
Very remote area	4.155* (2.449)	63.782	−0.046 (0.925)	1.047
Desired location (ref: foreign)				
Rural	0.914 (2.695)	2.493	2.571 (2.118)	13.083
Urban	−6.037** (2.688)	0.002	0.333 (1.180)	1.395
Group membership	2.649 (1.653)	14.141	0.926 (0.783)	2.524
Constant	−3.672 (3.869)	–	−4.075* (2.279)	–

Note: Cox and Snell R^2 (=0.698) and Nagelkerke R^2 (=0.788) suggest that a high share of the variations in the dependent variable are explained by the independent variables. Significance is indicated with ***$p<.01$, **$p<.05$, *$p<.1$. The exponentiated coefficients (Exp (B)) indicate to which extent a unit change in the explanatory variable changes the odds of falling into the group full-time (or part-time farming) while keeping all other variables constant.

Table 10. Parent-Related Factors Associated with Boys' Aspirations toward Agriculture

	Full-Time Farming		Part-Time Farming	
	Estimate (S.E)	Exp (B)	Estimate (S.E)	Exp (B)
Household farm size	0.313* (0.174)	1.368	0.221 (0.144)	1.248
Assets	−0.822*** (0.349)	0.440	−0.329 (0.288)	0.719
Farm income	−0.042 (0.040)	0.958	−0.061 (0.038)	0.941
Crop income	0.010 (0.024)	1.010	0.008 (0.023)	1.008
Father aspiration level	−0.065*** (0.025)	0.938	−0.036 (0.022)	0.965
Mother aspiration level	0.010 (0.019)	1.010	0.001 (0.017)	1.001
Power source (ref: own labor)				
Animal traction	21.887*** (1.066)	3,201	4.843** (2.437)	126.816
Hired labor	19.991 (0.000)	4,809	2.961 (2.188)	19.324
Father aspiration (ref: no farming)				
Full-time farming	0.038 (1.453)	1.038	−0.238 (1.225)	0.788
Part-time farming	0.396 (1.527)	1.485	−0.038 (1.240)	0.963
Mother aspiration (ref: no farming)				
Full-time farming	3.063** (1.332)	21.401	0.663 (1.132)	1.941
Part-time farming	−1.236 (1.519)	0.283	0.620 (1.199)	1.858
Constant	−16.497*** (4.233)	–	2.445 (4.618)	–

Note: Cox and Snell R^2 (=0.496) and Nagelkerke R^2 (=0.572) suggest that a high share of the variations in the dependent variable are explained by the independent variables. (***$p<.01$, **$p<.05$, *$p<.1$). The exponentiated coefficients (Exp (B)) indicate to which extent a unit change in the explanatory variable changes the odds of falling into the group full-time (or part-time farming) while keeping all other variables constant.

animal traction are much more likely to envision full-time or part-time farming and boys from households with more land are more likely to envision full-time farming relative to a non-farm livelihood. Boys whose mothers desire them to engage in full-time agriculture are more likely to envisage full-time farming. Male youth whose fathers have higher ambitions are less likely to envision full-time farming. Boys from households with more assets, an indicator of wealth, are more likely to plan to refrain from full-time farming.

Table 11 show that girls' aspirations to the farm are equally strongly associated with the main power source of farming according to the quantitative data from the household survey and that girls whose mothers aspire for them to farm are more likely to do so as well. Similar to the case of boys, fathers' general aspiration levels negatively correlate with girls' likelihood of envisaging a full-time farming career relative to a non-farm livelihood, but mothers' aspirations positively correlate with the girls' likelihoods to envision part-time farming.

Table 11. Parent-Related Factors Associated with Girls' Aspirations toward Agriculture

	Full-Time Farming		Part-Time Farming	
	Estimate (S.E)	Exp (B)	Estimate (S.E)	Exp (B)
Household farm size	0.154 (0.110)	1.167	0.076 (0.096)	1.079
Assets	−0.331 (0.244)	0.718	−0.270 (0.218)	0.763
Farm income	−0.008 (0.017)	0.992	−0.023 (0.016)	0.978
Crop income	0.001 (0.015)	1.001	0.019 (0.015)	1.019
Father aspiration level	−0.027* (0.015)	0.973	0.006 (0.014)	1.006
Mother aspiration level	0.014 (0.014)	1.014	0.026** (0.013)	1.026
Power source (ref: own labor)				
Animal traction	17.942*** (0.757)	6,194	0.208 (1.809)	1.232
Hired labor	17.324 (0.000)	3,341	−0.976 (1.790)	0.377
Father aspiration (ref: no farming)				
Full-time farming	0.138 (0.936)	1.148	−0.643 (0.924)	0.526
Part-time farming	−0.482 (0.895)	0.617	0.501 (0.756)	1.650
Mother aspiration (ref: no farming)				
Full-time farming	2.396*** (0.983)	10.979	1.064 (0.917)	2.897
Part-time farming	0.737 (0.917)	2.089	1.090 (0.750)	2.974
Constant	−18.080*** (1.964)	–	0.088 (2.707)	–

Note: Cox and Snell R^2 (=0.351) and Nagelkerke R^2 (=0.396) suggest that a high share of the variations in the dependent variable are explained by the independent variables. (***$p<.01$, **$p<.05$, *$p<.1$). The exponentiated coefficients (Exp (B)) indicate to which extent a unit change in the explanatory variable changes the odds of falling into the group full-time (or part-time farming) while keeping all other variables constant.

Which Actors Influence Youth Aspirations toward Agriculture?

Section "What are the Parents' Livelihood Aspirations for their Adolescent Sons and Daughters?" already gives some hints that the youth's aspirations toward farming are related to some degree to parents' general aspirations for themselves and their children. Figure 5 shows which actors influence youth livelihood aspirations, revealing that mothers and fathers are most influential according to the quantitative data from the household survey. Other notable actors are brothers, the schools, relatives, sisters, and friends, however, they all fall much behind the influence level of parents.

Figure 6 shows that fathers and mothers are mostly perceived to encouraged sons and daughters to engage in agriculture. In the case of girls, the share of parents discouraging them from farming or being neutral about a farming future was higher. The focus group discussions revealed that farm families often encourage their children, in particular sons, to farm by giving them some land to cultivate.

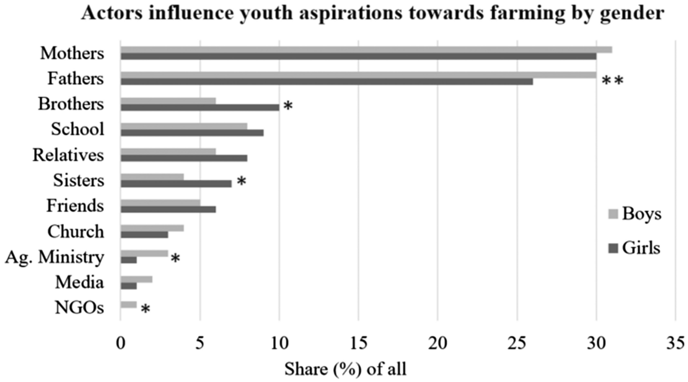

Figure 5. Actors' Influence on Youth Aspirations toward Farming by Gender. Asterisks denote significant differences according to Mann–Whitney tests with ***$p<.01$, **$p<.05$, *$p<.1$.

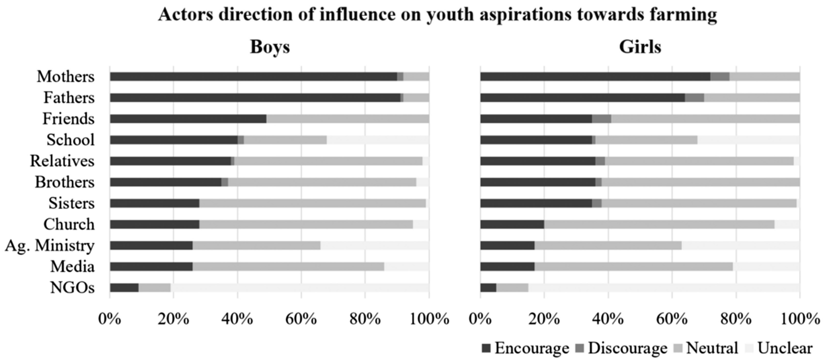

Figure 6. Actors' Direction of Influence on Youth Aspirations toward Farming.

> A child in a rural area cannot be educated without farming. (…) Our children, especially the male, are given a portion of land to cultivate while observing how family farmlands are cultivated. (Male adult FGD participant, Vubwi)

What Does the Perfect Farm Look Like?

Figure 7 shows how the youth imagined their "perfect farm" according to the data from the household survey. The youth's perceptions of the "perfect farm" allow for both insights into how farming has to look like for the youth to be more attracted as well as insights into what the youth are currently missing when it comes to farming; that

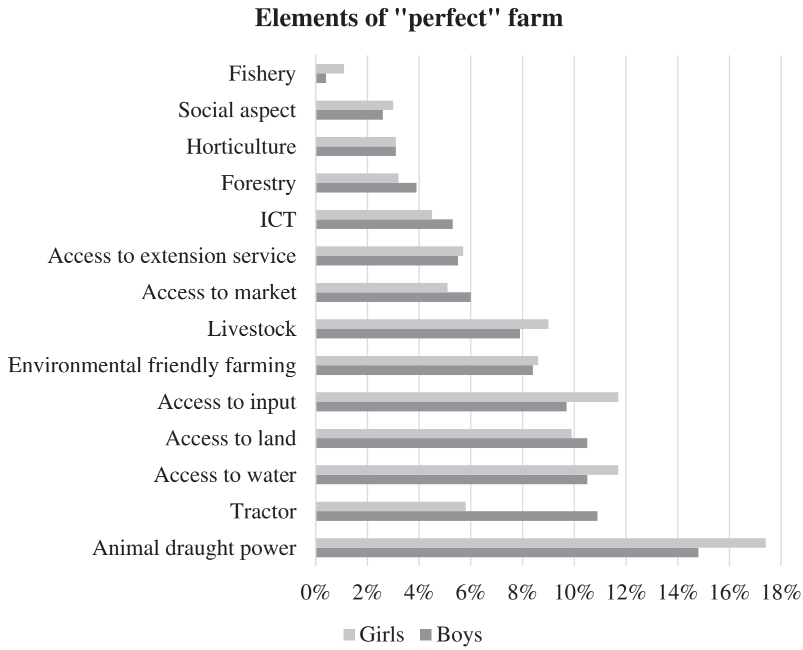

Figure 7. Key Elements of the Ideal Farm in Order of Importance by Gender.

is, factors that may affect their current satisfaction with farming, which greatly influences livelihood aspirations (see Tables 8 and 9). Figure 7 confirms the central role of labor burden and labor-saving mechanization. Both boys (15%) and girls (17%) saw animal draught power as a key element of the "perfect farm." 11% of the boys and 6% of the girls envisioned tractors to be the key power source on their farm, replacing manual labor. Other key elements include access to water, for example, in the form of irrigation, as well as inputs such as improved seeds, inorganic fertilizer and pesticides, and land. Many of these technologies can also save labor (e.g., irrigation using pumps and pesticides in the form of herbicides) and increase or safeguard yields (e.g., irrigation, and pesticides). Both boys and girls considered environmentally friendly farming to be key.

Discussion and Conclusion

This study has explored actors and factors shaping youth livelihood aspirations toward farming, in particular the role of parents' personal and farm characteristics, as well as structural gender differences such as those related to the access to land. A better understanding of the mechanisms affecting youth livelihood aspirations and gender differences can help to design more effective and appropriate policy interventions for "youth in agriculture."

The results reveal the large influence of parents' aspirations on children's livelihood aspirations. Fathers and mothers were perceived as (far) more influential in the formulation of the livelihood aspirations than other social network members such as peers, the church, or the media according to the analysis of the quantitative data from the household survey. The econometric results suggest a significant relationship between fathers' general aspirations and children's aspirations toward farming and between mothers' general and livelihood aspirations for their children and their likelihood of envisioning farming. This corresponds with Ball and Wiley (2005), who found a significant relationship between the aspirations of parents and children regarding the generational succession of family farms in the United States of America.

Parents mostly envisioned full-time farming for male youth according to the quantitative household survey, contradicting empirical research from other countries in Africa where most parents desired their children to leave farming (Tadele and Gella 2012; Verkaart et al. 2018). For daughters, parents typically envisioned livelihoods based on part-time farming or outside of agriculture. This corresponds with culturally prescribed gender roles such as male "breadwinners" and female "caregivers," as revealed by the qualitative focus group discussions. Fathers and mothers typically aspired to a "good marriage" for their daughters, which may also be influenced by the patrilineal system of land inheritance, giving women access to land mainly via their husbands. The more limited access to land for women may also thus explain the lower aspiration of the parents that their daughters become farmers.

Mirroring the parents' livelihood aspirations for them, the quantitative and qualitative results showed that boys are more inclined toward farming (full-time or part-time) than girls, who mostly desired part-time or non-farm livelihoods. This reflects several other studies from various developing countries (Elias et al. 2018; Rietveld et al. 2020; Tadele and Gella 2012). The qualitative data analysis suggests that the gender differences can be explained by the differences in the access to land between sons and daughters due to traditional norms, confirming findings from Bezu and Holden (2014) who found that access to land is key to understanding livelihood choices in Ethiopia. The gender differences can be explained by the parent's aspirations which are more likely to discourage daughters from farming (full-time) and encourage sons to farm according to both the quantitative and qualitative data. For sons who want to farm, this is positive reinforcement: they are supported to fulfill their livelihood aspirations, for example, by receiving farmland. In contrast, young men who do not want to farm may feel "locked" into farming by parental expectations and societal norms. This may help to explain why boys with high general aspiration levels are not more likely to refrain from farming than those with low general aspirations according to the quantitative analysis. In contrast, girls with higher general aspiration levels were significantly more likely to aspire to a future outside of farming according to the quantitative data. To some extent, this may also reflect gender norms; however, high-aspiring boys can also channel their energy into becoming large commercial farmers, while high aspiring girls face constraints to pursue such a vision and gender disparities often relegate women's roles as farmers to family subsistence according to the qualitative results. This would reflect Rietveld et al. (2020) who found that young women are particularly likely to see no future in commercial farming. For girls, this corresponds with the findings from a cross-country study where young people with high general aspirations were more commonly deserting rural areas and farming in search of high-status occupations and a better life (Mussa 2020).

The interest expressed by the youth in mixed livelihoods (45% of the males, 35% of the females) according to the household survey, with income streams from both farm and off-farm work, can also be understood as a risk-mitigating strategy (see also LaRue et al. 2021). Historical and contemporary factors such as the experiences of recurrent droughts since the 2010s and the economic crisis of the wider economy since the mid-2010s may show that both pure farming and non-farming livelihoods are associated with high risks, which can be mitigated to some degree with mixed livelihoods. Interestingly such mixed livelihoods were of particular interest for educated and high achieving girls according to the quantitative data. Distinguishing between full-time and part-time farming livelihoods hence provides an important improvement for understanding gendered farming aspirations as previous studies often presented farming as a take-it-or-leave-it option, neglecting the shades of gray between full-time farming and no farming (LaRue et al. 2021).

Next to parents' aspirations and support, farm satisfaction plays a large role. The quantitative results show a significant relationship between the youth's farm satisfaction and their aspirations toward farming. The observation that girls are—on average—less satisfied with their current farming experiences may also help to explain why they are more likely to envision a non-farming livelihood. Girls may be less satisfied because they face different tasks compared with boys and perceive farming as more burdensome (Rietveld et al. 2020). Moreover, while boys know they will obtain some farmland from their parents—and often already cultivate some farmland on their own—girls are less likely to receive farmland, which may affect their satisfaction level. However, the estimated share of farmland inherited was not significantly correlated with youth aspirations. Experiencing their mothers as less powerful regarding farm decision-making may also explain why girls are more likely to refrain from farming (Rietveld et al. 2020).

There is a need for continued effort to promote women's and girls' empowerment, both because this affects their well-being but also because this affects general and livelihood aspirations. The results show that gendered differences in the division of labor and access to and control over resources can influence the scope and range of young people's livelihood aspirations. Gendered disparities such as the intergenerational transfer of land from fathers to sons align with the cultural expectation of women in caregiving roles (as wives to male landowners, or as nurses/teachers) and men in leadership and provisioning roles (as husbands and household heads, or as doctors) as suggested from the qualitative results. While the study indicates parents' interest in their children's welfare, the unequal opportunities—manifested in the forms of gendered socialization and entitlement—inevitably translate into gendered aspirations for their children (Quisumbing 2007). These differences in treatment and expectation significantly encourage and sometimes curtail the range of livelihood aspirations, particularly for young girls, as clearly shown in the qualitative data analysis.

Efforts to reduce the barriers faced by girls to aspire to occupations of "choice" are, therefore, encouraged. Such initiatives may involve engaging men and boys as positive agents of change through exchanges with women and girls to harness opportunities that challenge embedded beliefs and gender barriers. If mothers faced fewer burdens in farming, for example, in the form of increased access to inputs and/or more equal control over the use of farming incomes, perhaps more young girls would aspire to be commercial or full-time farmers. It follows that rural and agricultural transformation can also affect boys' and girls' aspirations. It is also noteworthy that the "perfect farm" mostly centered on labor-saving mechanization (Daum 2019). The regression analysis also suggests a strong link between the types of power sources on the parents' farms and aspirations, highlighting the need for continued efforts for mechanization to make farming attractive (Daum and Birner 2020). Beyond mechanization, several other aspects are central to the "perfect farm," including not only access to water, land, inputs, and services but also the notion of "environmentally friendly farming."

Overall, the strong link between parents' and children's livelihood aspirations, and the key role of parents in the realization of children's livelihood aspirations (e.g., through the transfer of land), suggest that policies, programs, and projects focusing on rural youth should go beyond working with the youth alone. Instead, a whole-family approach seems recommendable, where both parents and youth are equally addressed. Such an approach, bringing parents on board, could facilitate a reflection on prevailing gender norms, roles, and stereotypes and open the opportunity spaces for youth. This can help avoid the negative outcomes of a more exclusive approach. For example, widening youth livelihood aspirations but not opportunity spaces, which parents partly determine, could lead to conflicts and frustration, which could be minimized with such an approach. For example, as shown above, many families (and the wider rural communities) hold the age-long tradition of favoring males in the sharing of inheritance (i.e., land) to the detriment of females who might be interested in farm-related livelihoods. Advocacy campaigns and sensitization could lead to more opportunities for young people willing to engage in farming regardless of gender or status in society. Regarding non-farm jobs, the whole-family approach could be implemented, for example, in the form of parent-teacher associations or locally organized "career fairs" for youth and their parents. The "whole-family" approach can benefit from insights from gender studies, which have emphasized the need to consider household dynamics and avoid seeing empowerment as a zero-sum game (Salia et al. 2018). Without such initiatives, efforts to encourage specific livelihood pathways for boys and girls may be undermined by conflicting parents' aspirations, leaving the youth torn in-between. This may be of particular importance where parents are unable to assess the "opportunity space" of the youth. The strong role of parents' aspirations raises some doubts on whether youth can exercise agency in pursuit of their "true" aspirations. For example, boys may pursue a farming career to conform to their parents' livelihood aspirations and societal norms, and, vice versa, girls may pursue a non-farming career to also conform to their parent's aspirations and social norms. This tension between obligations and livelihood aspirations may threaten the youth's future well-being.

This study faces some limitations, including small sample size and external validity. Both "parents and youth" aspirations may reflect localized socio-economics factors such as culture, traditions, social norms, and the "opportunity space" (IFAD 2019). As shown by Chamberlin et al. (2021), rural vibrancy affects youth aspirations to some degree. All these aspects may be different in other African countries, as well as in other low- to middle-income countries. This also includes the importance of parents for the formation of youth aspirations, which may be more or less important in other countries. However, as shown by the World Value Survey (2020), "traditional societies," which are characterized by a strong respect for one's parents ("regardless of how they behave"), are common across African, Arab, and Latin American countries. In such societies, the suggested "whole-family" approach seems to be of particular importance to ensure that youth can pursue livelihoods that can sustain them—physically and psychologically.

References

Adu-Baffour, Ferdinand, Thomas Daum, and Regina Birner. 2019. "Can Small Farms Benefit from Big Companies' Initiatives to Promote Mechanization in Africa? A Case Study from Zambia." *Food Policy* 84:133–45.

Ali, Daniel Ayalew, Klaus Deininger, Dorothea Huberta Maria Hilhorst, Frank Kakungu, and Yuanyuan Yi. 2019. *Making Secure Land Tenure Count for Global Development Goals and National Policy: Evidence from Zambia*. World Bank Policy Research Working Paper 8912.

Amberntsson, Pelle. 2020. "The Past of the Present: Exploring Contemporary Rural Livelihoods from A Historical Perspective—The Case of Smallholders in Eastern Zambia." *South African Historical Journal* 72(4):651–76.

Anyidoho, Nana Akua, Jennifer Leavy, and Kwadwo Asenso-Okyere. 2012. "Perceptions and Aspirations: A Case Study of Young People in Ghana's Cocoa Sector." *IDS Bulletin* 43(6):20–32.

Bajema, Duane, W. Wade Miller, and David L. Williams. 2002. "Aspirations of Rural Youth." *Journal of Agricultural Education* 43(3):61–71.

Ball, Anna L. and Angela Wiley. 2005. "The Aspirations of Farm Parents and Pre-Adolescent Children for Generational Succession of the Family Farm." *Journal of Agricultural Education* 46(2):36–46.

Bernard, Tanguy, Stefan Dercon, Kate Orkin, and Alemayehu Taffesse. 2014. *The Future in Mind: Aspirations and Forward-Looking Behaviour in Rural Ethiopia*. London: Centre for Economic Policy Research.

Bernard, Tanguy and Alemayehu Taffesse. 2014. "Aspirations: An Approach to Measurement with Validation Using Ethiopian Data." *Journal of African Economies* 23(2):189–224.

Bessant, Judith. 2018. "Young Precariat and a New Work Order? A Case for Historical Sociology." *Journal of Youth Studies* 21(6):780–98.

Bezu, Sosina and Stein Holden. 2014. "Are Rural Youth in Ethiopia Abandoning Agriculture?" *World Development* 64:259–72.

von Braun, Joachim and Chiara Kofol. 2017. *Expanding Youth Employment in the Arab Region and Africa*. ZEF-Center for Development Research, University of Bonn, Working Paper, 155.

Bullock, Renee and Todd Crane. 2021. "Young Women's and Men's Opportunity Spaces in Dairy Intensification in Kenya." *Rural Sociology* 86(4):777–808.

Byun, Soo-yong, Judith L. Meece, Matthew J. Irvin, and Bryan C. Hutchins. 2012. "The Role of Social Capital in Educational Aspirations of Rural Youth." *Rural Sociology* 77(3):355–79.

Chamberlin, Jordan, Cristina Ramos, and Kibrom Abay. 2021. "Do More Vibrant Rural Areas have Lower Rates of Youth Out-Migration? Evidence from Zambia." *The European Journal of Development Research* 33(4):951–79.

Chapoto, Antony, Thomas Jayne, and Nicole M. Mason. 2011. "Widows' Land Security in the Era of HIV/AIDS: Panel Survey Evidence from Zambia." *Economic Development and Cultural Change* 59(3):511–47.

Daum, Thomas. 2019. "Of Bulls and Bulbs: Aspirations, Opinions and Perceptions of Rural Adolescents and Youth in Zambia." *Development in Practice* 29(7):882–97.

Daum, Thomas and Regina Birner. 2020. "Agricultural Mechanization in Africa: Myths, Realities and An Emerging Research Agenda." *Global Food Security* 26:100393.

———. 2021. "The Forgotten Agriculture-Nutrition Link: Farm Technologies and Human Energy Requirements." *Food Security* 14(2):395–409.

Daum, Thomas, Filippo Capezzone, and Regina Birner. 2021. "Using Smartphone App Collected Data to Explore the Link between Mechanization and Intra-Household Allocation of Time in Zambia." *Agriculture and Human Values* 38(2):411–29.

Daum, Thomas, Alysha Vehre, Carolin Schweizerhof, and Christoph Schunko. 2022. "Edible Weeds and Food and Nutrition Security in the Face of the Herbicide Revolution. A Case Study from Zambia." *International Journal of Agricultural Sustainability*: 1–15. https://doi.org/10.1080/14735903.2022.2065949

Davis, Benjamin, Stefania Di Giuseppe, and Alberto Zezza. 2017. "Are African Households (Not) Leaving Agriculture? Patterns of Households' Income Sources in Rural Sub-Saharan Africa." *Food Policy* 67:153–74.

Doss, Cheryl, Ruth Meinzen-Dick, Agnes Quisumbing, and Sophie Theis. 2018. "Women in Agriculture: Four Myths." *Global Food Security* 16:69–74.

Elias, Marlène, Netsayi N. Mudege, Diana E. Lopez, Dina Najjar, Vongai Kandiwa, Joyce S. Luis, Jummai Yila, Amare Tegbaru, Gaya Ibrahim, Lone Badstue, Esther Njuguna-Mungai, and Abderahim Bentabu. 2018. "Gendered Aspirations and Occupations among Rural Youth, in Agriculture and Beyond: A Cross-Regional Prescriptive." *Journal of Gender, Agriculture and Food Security* 3(1):82–107.

FAO, IFAD, and WFP. 2015. *Developing the Knowledge, Skills and Talent of Youth To Further Food Security*. Rome: Food and Agriculture Organization.

von Grebmer, Klaus, Jill Bernstein, Fraser Patterson, Miriam Wiemers, Réiseal Ní Chéilleachair, Connell Foley, Seth Gitter, Kierstin Ekstrom, and Heidi Fritschel. 2019. *Global Hunger Index 2019: The Challenge of Hunger and Climate Change*. Bonn and Dublin: Welthungerhilfe and Concern International.

Gumbo, Davison J., Kondwani Y. Mumba, Moka M. Kaliwile, Kaala B. Moombe, and Tiza I. Mfuni. 2016. "Agrarian Changes in the Nyimba District of Zambia." in *Agrarian Change in Tropical Landscapes*, edited by L. Deakin, M. Kshatriya, and T. Sunderland. Bogor: CIFOR.

Gutman, Leslie and Rodie Akerman. 2008. *Determinants of Aspirations*. London: Centre for Research on the Wider Benefits of Learning.

Huijsmans, Roy, Aprilia Ambarwati, Charina Chazali, and M. Vijayabaskar. 2021. "Farming, Gender and Aspirations Across Young People's Life Course: Attempting to Keep Things Open While Becoming a Farmer." *The European Journal of Development Research* 33(1):71–88.

IAPRI. 2016. *Rural Agricultural Livelihoods Survey: 2015 Survey Report*. Lusaka: Indaba Agricultural Policy Research Institute (IAPRI).

———. 2019. *Rural Agricultural Livelihoods Survey: 2019 Survey Report*. Lusaka: Indaba Agricultural Policy Research Institute.

Jungen, Kristen Anne. 2008. *Parents Influence and Career Choice: How Parents Affect the Career Aspirations of their Children*. Menomonie, WI: University of Wisconsin-Stout.

Kanduza, Ackson. 1987. "Allocation System of Land and its Guardianship Under Traditional Tenure in Zambia." *Journal of Eastern African Research & Development* 17:1–17.

———. 1992. *Socio-Economic Change in Eastern Zambia: Pre-Colonial to the 1980s*. Lusaka: Historical Association of Zambia.

LaRue, Katie, Thomas Daum, Kai Mausch, and David Harris. 2021. "Who Wants to Farm? Answers Depend on How You Ask: A Case Study on Youth Aspirations in Kenya." *The European Journal of Development Research* 33(4):1–25.
Leavy, Jennifer and Naomi Hossain. 2014. *Who Wants to Farm? Youth Aspirations, Opportunities and Rising Food Prices.* IDS Working Papers 439:1–44.
Leavy, Jennifer and Sally Smith. 2010. *Future Farmers: Youth Aspirations, Expectations and Life Choices.* Future Agricultures Discussion Paper 13. Sussex: Institute of Development Studies.
Lubungu, Mary and Regina Birner. 2021. "Gender Relations in Smallholder Cattle Production in Zambia." *World Development Perspectives* 22:100309.
Lungkang, Dina Magdalena. 2018. *Shaping Aspirations: Insights of Young Farmers Life Trajectory in Pinrang Regency.* The Hague: International Institute of Social Studies.
Mason, Nicole M., Ayala Wineman, and Solomon T. Tembo. 2020. "Reducing Poverty by 'Ignoring the Experts'? Evidence on Input Subsidies in Zambia." *Food Security* 12(5):1157–72.
Mulenga, Brian P., Hambulo Ngoma, and Solomon Tembo. 2015. "Climate Change and Agriculture in Zambia: Impacts, Adaptation and Mitigation options." In *Agriculture in Zambia: Past, Present, and Future*, edited by A. Chapoto and N. J. Sitko. Lusaka: Indaba Agricultural Policy Research Institute.
Mussa, Essa Chanie. 2020. *Youth Aspirations, Perceptions of Farming, and Migration Decisions in Rural Sub-Saharan Africa: Further Empirical Evidence from Ethiopia.* ZEF Discussion Paper 296. Bonn: Center of Development Research (ZEF).
Muwi, Lynn Rutendo. 2012. *Rural Youth and Smallholder Farming: The Present and Future of Agrarian Activities from Generational Perspectives.* The Hague: International Institute of Social Studies.
Paloş, Ramona and Loredana Drobot. 2010. "The Impact of Family Influence on the Career Choice of Adolescents." *Procedia-Social and Behavioral Sciences* 2(2):3407–11.
Pelekamoyo, Joan and Bridget Bwalya Umar. 2019. "Access to and Control Over Agricultural Labor and Income in Smallholder Farming Households: A Gendered Look from Chipata, Eastern Zambia." *Journal of Gender, Agriculture and Food Security* 4(2):42–57.
Peng, Chao-Ying Joanne, Tak-Shing Harry So, Frances K. Stage, and Edward P. St John. 2002. "The Use and Interpretation of Logistic Regression in Higher Education Journals: 1988–1999." *Research in Higher Education* 43(3):259–93.
Peng, Chao-Ying Joanne, and Rebecca Naegle Nichols. 2003. "Using Multinomial Logistic Models to Predict Adolescent Behavioral Risk." *Journal of Modern Applied Statistical Methods* 2(1):177–88.
Pereznieto, Paola, Rachel Marcus, Matthew Maclure, Nandini Archer, Anna Mdee, Karen Moore, and Rose A. Miller. 2018. *Gender and Youth Livelihoods Programming in Africa.* London: Overseas Development Institute and Mastercard Foundation.
Quisumbing, Agnes R. 2007. *Investments, Bequests, and Public Policy: Intergenerational Transfers and the Escape from Poverty.* Chronic Poverty Research Centre Working Paper 98. Washington, DC: Chronic Poverty Research Centre.
Quisumbing, Agnes R., Ruth Meinzen-Dick, Terri L. Raney, André Croppenstedt, Julia A. Behrman, and Amber Peterman. 2014. *Gender in Agriculture: Closing the Knowledge Gap.* Dordrecht: Springer.
Rietveld, Anne M., Margreet van der Burg, and Jeroen C. J. Groot. 2020. "Bridging Youth and Gender Studies to Analyse Rural Young Women and Men's Livelihood Pathways in Central Uganda." *Journal of Rural Studies* 75:152–63.
Ripoll, Santiago, Jens Andersson, Lone Badstue, Marion Büttner, Jordan Chamberlin, Olaf Erenstein, and James Sumberg. 2017. "Rural Transformation, Cereals and Youth in Africa: What Role for International Agricultural Research?" *Outlook on Agriculture* 46(3):168–77.
Salia, Samuel, Javed Hussain, Ishmael Tingbani, and Oluwaseun Kolade. 2018. "Is Women Empowerment a Zero Sum Game? Unintended Consequences of Microfinance for

Women's Empowerment in Ghana." *International Journal of Entrepreneurial Behavior & Research* 24(1):273–89.

Sherwood, Richard, Jr. 1989. "A Conceptual Framework for the Study of Aspirations." *Research in Rural Education* 6(2):61–6.

Sitko, Nicholas J. and Jordan Chamberlin. 2016. "The Geography of Zambia's Customary Land: Assessing the Prospects for Smallholder Development." *Land Use Policy* 55:49–60.

Smale, Melinda, Veronique Theriault, and Hamza Haider. 2019. "Intrahousehold Productivity Differentials and Land Quality in The Sudan Savanna of Mali." *Land Economics* 95(1):54–70.

Sumberg, James, Thomas Yeboah, Justin Flynn, and Nana Akua Anyidoho. 2017. "Young People's Perspectives on Farming in Ghana: A Q Study." *Food Security* 9(1):151–61.

Tadele, Getnet and Asrat Ayalew Gella. 2012. "'A Last Resort and Often Not an Option at All': Farming and Young People in Ethiopia." *IDS Bulletin* 43(6):33–43.

Üllenberg, Alfons, Margitta Minah, Theo Rauch, and Daniela Richter. 2017. *Zambia: Towards Inclusive and Sustainable Rural Transformation.* Albrecht Daniel Thaer-Institut für Agrar-und Gartenbauwissenschaften.

UNICEF. 2021. *Children in Zambia.* United Nations Children's Fund. Retrieved December 21, 2021 (https://www.unicef.org/zambia/children-zambia).

Vail, Leroy. 1977. "Ecology and History: The Example of Eastern Zambia." *Journal of Southern African Studies* 3(2):129–55.

Verkaart, Simone, Kai Mausch, and David Harris. 2018. "Who Are Those People We Call Farmers? Rural Kenyan Aspirations and Realities." *Development in Practice* 28(4):468–79.

Wang, Donghui, Annelise DeJong Hagedorn, and Diane K. McLaughlin. 2021. "Patterns of Educational, Occupational, and Residential Aspirations of Rural Youth: The Role of Family, School, and Community." *Rural Sociology* 86(2):177–203.

Wiley, Angela R., Timothy Bogg, and Moon-Ho Ringo Ho. 2005. "The Influence of Parental Socialization Factors on Family Farming Plans of Preadolescent Children: An Exploratory Analysis." *Journal of Research in Rural Education* 20(11):12.

World Bank. 2021a. *World Bank Open Data.* World Bank. Retrieved December 21, 20121 (https://data.worldbank.org/).

———. 2021b. *The World Bank in Zambia—Overview.* World Bank. Retrieved December 21, 2021 (https://www.worldbank.org/en/country/zambia/overview#1).

World Value Survey. 2020. *The Inglehart-Welzel World Cultural Map—World Values Survey 7 (2020).* Retrieved April 8, 2021 (https://www.worldvaluessurvey.org/WVSContents.jsp?CMSID=Findings).

Yeboah, Thomas, James Sumberg, Justin Flynn, and Nana Akua Anyidoho. 2017. "Perspectives on Desirable Work: Findings from a Q Study with Students and Parents in Rural Ghana." *The European Journal of Development Research* 29(2):423–40.